A cubic weakly reversible mass-action system with an algebraic limit cycle

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The question

Craciun and Erban [1] introduce chemical analogues of Hilbert's 16th problem. For $n \in \mathbb{N}$, W_n is the set of two-species mass-action systems whose reactions have at most n reactant and at most n product molecules (the set M_n of [1, Definition 2]) and whose reaction graph is weakly reversible [1, Definition 3]. $W^a(n)$ is the maximum number of algebraic limit cycles of systems in W_n [1, Definition 5]. Their Table 2 records $W^a(2) = 0$, $W^a(3) \geq 0$ and $W^a(4) \geq 1$, and they write (arXiv:2406.05057v2, §4 and §6):

> "The question of whether cubic weakly reversible two-species systems can give rise to algebraic limit cycles remains open, leaving us with inequality $W^{(a)}(3) \geq 0$ in Table 2."
> "[. . .] it remains an open question whether a cubic weakly reversible system can have an algebraic limit cycle."

Theorem. $W^a(3) \geq 1$. Explicitly, the weakly reversible network with integer rate constants

$$0 \xrightarrow{434} 2Y, \qquad 0 \xrightarrow{448} X+Y, \qquad Y \xrightarrow{3374} 0, \qquad 2Y \xrightarrow{1728} 3Y,$$
$$3Y \xrightarrow{194} Y, \qquad 3Y \xrightarrow{11} 2X+Y, \qquad X+Y \xrightarrow{328} X+2Y,$$
$$X+2Y \xrightarrow{250} 3Y, \qquad 2X+Y \xrightarrow{216} 3X, \qquad 3X \xrightarrow{28} X+2Y$$

belongs to W_3, and its mass-action system

$$\dot{x} = f(x,y) = -56x^3 + 216x^2y - 250xy^2 + 22y^3 + 448,$$
$$\dot{y} = g(x,y) = 56x^3 - 216x^2y + 250xy^2 + 328xy - 410y^3 + 1728y^2 - 3374y + 1316$$

has the circle $\mathcal{C} = \{(x-2)^2+(y-2)^2 = 1\} \subset \mathbb{R}^2_{>0}$ as a hyperbolic, orbitally asymptotically stable limit cycle.

1 Proof

1. The network is in W_3. Its eight complexes $0, Y, 2Y, 3Y, X+Y, X+2Y, 2X+Y, 3X$ have molecularity at most 3, so every reaction has at most 3 reactant and 3 product molecules. The reaction graph is strongly connected: $0 \to 2Y \to 3Y \to Y \to 0$, $0 \to X+Y \to X+2Y \to 3Y$, and $3Y \to 2X+Y \to 3X \to X+2Y \to 3Y$ are directed cycles whose union contains every reaction and passes through every complex. Hence every reaction lies on a directed cycle: the network is weakly reversible (it has a single linkage class). The mass-action equations follow from $\dot{z} = \sum_{\text{reactions}} \kappa\,(\text{product} - \text{source})\, x^a y^b$.

2. The circle is invariant. With $h(x,y) = (x-2)^2 + (y-2)^2 - 1$, direct expansion gives the polynomial identity

$$f\,h_x + g\,h_y = s\,h, \qquad s(x,y) = -112x^2 + 544xy - 448x - 820y^2 + 1728y - 1008.$$

So $\dot{h} = s\,h$ along solutions, and $\mathcal{C} = \{h = 0\}$ is invariant. It lies in $[1,3]^2$, inside the open positive quadrant.

3. There is no equilibrium on $\mathcal{C}$. Parametrise $\mathcal{C}$ by $x = 2 + \cos\theta$, $y = 2 + \sin\theta$. On $\mathcal{C}$, the field is tangent, so $F = (f,g) = P\,(-\sin\theta, \cos\theta)$ with $P = -(y-2)f + (x-2)g$, and $\dot{\theta} = P$. We show $P > 0$ on $\mathcal{C}$. The resultant $\mathrm{Res}_y(P,h) \in \mathbb{Q}[x]$ (degree 4; the leading coefficient of h in y is 1) has no real root in $[1,3]$ (Sturm count), so P and h have no common real zero, i.e. $P \neq 0$ on $\mathcal{C}$; since $P(3,2) = 792 > 0$ and $\mathcal{C}$ is connected, $P > 0$ on $\mathcal{C}$. Independently, a ball-arithmetic evaluation on 4096 subintervals of $[0, 2\pi]$ gives $P \geq 348$ on $\mathcal{C}$.

Consequently $\mathcal{C}$ is a periodic orbit, traversed counter-clockwise, of period $T = \int_0^{2\pi} d\theta / P(\theta) \in [0.0103320434082921 \pm 4.3 \times 10^{-17}]$.

4. The periodic orbit is hyperbolic and stable. For a periodic orbit γ of a planar C^1 vector field, the derivative of the Poincaré return map at γ equals $\exp\!\left(\oint_\gamma \operatorname{div} F\,dt\right)$ (e.g. Perko [3, §3.4, Theorem 2]). Here

$$\oint_{\mathcal{C}} \operatorname{div} F\,dt = \int_0^{2\pi} \frac{\operatorname{div} F(2+\cos\theta, 2+\sin\theta)}{P(\theta)}\,d\theta \in [-3.68475192932676 \pm 3.3 \times 10^{-15}],$$

computed by rigorous ball-arithmetic integration (Arb's `acb_calc_integrate` [4] via python-flint, which returns a guaranteed enclosure for integrands analytic near the path; P has no zero near $[0, 2\pi]$ by step 3). The integral is negative, so the return map is a contraction at γ: $\mathcal{C}$ is a hyperbolic, orbitally asymptotically stable periodic orbit, in particular isolated among periodic orbits, i.e. a limit cycle. It is algebraic, being a component of $\{h = 0\}$. As a consistency check, $\oint_{\mathcal{C}} s\,dt$ has the same enclosure (the coordinate h is transversal to $\mathcal{C}$ and satisfies $\dot{h} = sh$, so both integrals give the transversal Floquet exponent). □

2 Remarks

1. **Construction.** With h fixed, the conditions "$fh_x + gh_y = sh$" and "the realisation uses only complexes of molecularity ≤ 3" are linear in the rate constants and in the coefficients of s; weak reversibility is enforced by requiring a strictly positive circulation on the support of the rate vector. A linear program found a 17-reaction network during the exploration phase (also certified, see below); a mixed-integer program minimising the number of reactions (SCIP) then returned the 10-reaction support above, on which the solution of the linear identity is unique up to a time rescaling. This gives the integer rate constants.
2. **A second example.** The 17-reaction network listed in `scripts/certify_cycle.py` (network N17; 10 complexes, rational rate constants) also has $\mathcal{C}$ as a stable hyperbolic limit cycle, with $\oint \operatorname{div} F\,dt \in [-0.636539668381478 \pm 2 \times 10^{-16}]$ and period $1.91598\ldots$.
3. **Consequences for the table of [1].** Since $W_3 \subseteq M_3$ and $W_3 \subseteq W_4$, the example also re-proves $M^a(3) \geq 1$ and $W^a(4) \geq 1$ with a cubic system.

4. **Deficiency.** The network has 8 complexes, one linkage class and rank 2, hence deficiency 5; it is not covered by the deficiency-zero or deficiency-one theorems, consistent with the presence of a limit cycle.

3 Verification

`scripts/certify_cycle.py` checks, for both networks, in exact rational arithmetic (sympy): membership in W_3 (molecularities, strong connectivity), the invariance identity with its cofactor, the resultant and the sign of P; and in rigorous ball arithmetic (python-flint/Arb): the positivity of P on 4096 subintervals, and enclosures of the period, of $\oint s\,dt$ and of $\oint \operatorname{div} F\,dt$. Log: `certificates/certify_cycle.log` (all PASS, about 3 s).

References

1. G. Craciun, R. Erban, *Planar chemical reaction systems with algebraic and non-algebraic limit cycles*, J. Math. Biol. (2025), doi:10.1007/s00285-025-02221-0; arXiv:2406.05057v2.
2. M. Feinberg, *Foundations of Chemical Reaction Network Theory*, Springer (2019) — weak reversibility, deficiency.
3. L. Perko, *Differential Equations and Dynamical Systems*, 3rd ed., Springer (2001), §3.4.
4. F. Johansson, *Numerical integration in arbitrary-precision ball arithmetic*, in: Mathematical Software – ICMS 2018, LNCS 10931, Springer (2018) 255–263, arXiv:1802.07942.